

JOSEMARIA Y. LAYUG III

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PROFILE

Computer Science graduate (BS and MS, Ateneo de Manila, Magna Cum Laude) with a research background in real-time audio signal processing, using Python, C++, PyTorch, and JUCE, and a paper accepted at ECTI DAMT & NCON 2026. Eight years of leadership and administrative experience, including three years running an independent freelance business managing logistics, client relations, and event operations for media production for clients including UP and Ateneo student organizations, Sony Music artists, and large independent creators. Builds software independently, including a personal portfolio site coded from scratch in HTML, CSS, and JavaScript. Comfortable across AI-assisted development workflows, general software development, and multimedia systems, and looking to bring that mix of technical depth and business management experience to a broader engineering role.

TECHNICAL SKILLS

Productivity: Google Workspace, Notion, Zapier, Microsoft 365, Apps Script, Zoom

Languages: Python, Java, C++, SQL, HTML/CSS, JavaScript, Google Apps Script

Systems: Linux (Ubuntu), networking basics (Jellyfin, Deluge, Cockpit)

Media Tools: Logic Pro, OBS Studio, DaVinci Resolve, Figma, Final Cut Pro, Adobe Premiere Pro, After Effects, Photoshop, Illustrator

EDUCATION

Ateneo de Manila University — Master of Science in Computer Science

Jun. 2024 – Jun. 2026

- Conference Publication: 2026 ECTI DAMT & NCON — AI and Machine Learning
- Thesis: "Real-Time Speaker-Sensitive Audio Gating via Supervised Frame-Level Speaker Recognition"

Ateneo de Manila University — BS Computer Science, Interactive Multimedia

Jul. 2020 – Jun. 2024

- Graduated Magna Cum Laude, Grade: 3.74
- Thesis: "An Analysis of Queen's Discography Using K-Means, DBSCAN, and Affinity Propagation Clustering Techniques"

TECHNICAL PROJECTS

Portfolio Website

Personal

- Built a personal portfolio site from scratch in vanilla HTML, CSS, and JavaScript, featuring tab switchers, accordion cards, a media modal for Spotify/YouTube embeds, a sticky YouTube miniplayer, lazy-loaded thumbnails, and a responsive hamburger menu; deployed on GitHub Pages with a custom domain configured via Namecheap DNS.

Real-Time Speaker-Sensitive Audio Gating via Supervised Frame-Level Speaker Recognition

Ateneo de Manila University

- Trained a supervised deep learning model in Python on Google Colab using noise injected speech samples. Applied data augmentation and performed a comparative analysis with two multistyle training methodologies to improve model robustness.
- Implemented and tested on Logic Pro as a real-time audio plug-in in C++ via the JUCE framework.
- Publication: 2026 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunication Engineering (ECTI DAMT & NCON)

Automated Video Captioning Pipeline (Python)

RausHaus Productions

- Built an end-to-end Python pipeline for karaoke-style subtitles: isolated vocal audio is fed into stable-ts (OpenAI Whisper-based) for forced lyric alignment, output to .ass, verified in Aegisub, then converted to .ytt via YTSUBConverter for YouTube publishing

Host OS Impact on Type 2 Hypervisor Performance (Research)

Ateneo de Manila University

- Co-authored a systems research paper benchmarking Windows 11, macOS, and Linux Mint as host operating systems for a VirtualBox Type 2 hypervisor, using a triple-boot test environment (Boot Camp and dual-boot)
- Designed the benchmarking methodology across Geekbench, PassMark, RAMSpeed, and MPrime to evaluate CPU/RAM performance, thermal behavior, and resource utilization on a virtualized Linux guest OS

Gilas Pilipinas Fan Discourse Analysis (Python)

Ateneo de Manila University

- Built a data pipeline scraping and analyzing fan sentiment: BeautifulSoup and the YouTube Data API for collection, NLTK for text preprocessing and sentiment classification, and scikit-learn for LDA topic modeling

Real-Time Speaker-Sensitive Audio Gating via Supervised Frame-Level Speaker Recognition

Ateneo de Manila University

- Extracted musical features from MIDI data (Lakh MIDI Dataset, via jSymbolic) and applied comparative clustering analysis across full and reduced feature sets
- Built custom Python and Google Sheet tooling for cluster analysis and Spotify playlist generation (Spotipy)

EXPERIENCE

RausHaus Productions

Feb. 2020 – Present

Founder

- Manage an independent music studio, overseeing music and video production, logistics, finances, and distribution operations
- Execute events end-to-end with local artists, venues, and sponsors, coordinating budgets, contracts, and schedules

Event Manager

- Delivered 8+ live productions and album launch events (2022–2025), managing full production lifecycles from planning through execution

Ateneo Musicians' Pool

Apr. 2021 – Jun. 2024

Executive Vice President for Administration (Apr. 2023 – Jun. 2024)

- Developed and oversaw the application process for 600+ applicants; directed administrative operations and streamlined processes across departments
- Managed organizational databases, digital workspaces, and document systems; led officer transition and knowledge transfer

Artist Management Head (Jun. 2022 – Apr. 2023)

- Designed and implemented an application and evaluation process for 250+ applicants; oversaw deployment of 170+ members

Sound Engineering Head / Project Manager (Apr. 2021 – Jun. 2022)

- Facilitated technical training and project deployment for sound engineers

Project Manager (Apr. 2021 – Apr. 2023)

- Initiative Head - TONEEJAY x Jess & Pat's Workshop (2023)
- Coordinator - Rites of Passage (2022)
- Initiative Head - AMazing (2021)
- Lead Sound Engineer - cAMP (2022)
- Logistics Head - Rites of Passage (2021)

Kroma Entertainment Inc.

Jun. – Jul. 2023

Strategy, Project Management, and Market Research

- Created process mapping diagrams for intra-business units; produced market research, strategy analyses, and presentations for 3 business sub-units

VOLUNTEER WORK

- Technicals Head, KlimAksyon Environmental Awareness Webinar (Kalinga, 2020) — trained a livestreaming team and handled technical production for a 2-day webinar
- Stream Technical, eSports Tournament (Kinabookasan Literacy Program, 2020) — built scenes and managed tournament livestream
- Productions Head, Paralaya: Yugto Benefit Concert (2019–2020) — led a 3,000+ attendee benefit concert, managing budget, logistics, and artist bookings
- Productions Head, Pr1mero: Sulyap Benefit Concert (2018–2019) — led a 2,000+ attendee benefit concert, negotiating and securing budget, logistics, and agreements for artist bookings